

Runsheng Liu

liurunsheng@pku.edu.cn

PERSONAL INFORMATION

D.O.B.: 02 March, 2000

Nationality: China

EDUCATION

Peking University 2022–2027 (expected)

Ph.D. student in Probability and Statistics

Advisor: Prof. Xinyi Li

Peking University 2018–2022

B.S. in Mathematics

RESEARCH INTERESTS

Probability theory with focus on random fractals.

WORKS

1. **The two-sided multifractal spectrum of Schramm-Loewner evolution**, with Gefei Cai and Yifan Gao. *In preparation*.
2. **An explicit lower bound for the growth exponent of three-dimensional loop-erased random walk**, *Preprint*, available at arXiv:2609.18643.
3. **Tail exponents of the three-dimensional uniform spanning tree and Abelian sandpile**, with Xinyi Li and Daisuke Shiraishi. *Preprint*, available at arXiv:2605.19419.
4. **Non-existence of several random fractals in Brownian motion and Brownian loop soup**, with Yifan Gao, Xinyi Li and Wei Qian. To appear in *Ann. Probab.*, also available at arXiv:2507.20324.
5. **Convergence in natural parametrization of random walk frontier**, with Yifan Gao, Xinyi Li, Xiangyi Liu and Daisuke Shiraishi. *Sci. China Math.*, 2026, 69(5): 1299–1330.
6. **A boundary Harnack principle and its application to analyticity of 3D Brownian intersection exponents**, with Yifan Gao, Xinyi Li, Yifan Li and Xiangyi Liu. *Preprint*, available at arXiv:2411.14921.
7. **The intermediate level-sets of the four-dimensional membrane model**, with Xinyi Li. *Acta Math. Sinica, Eng. Ser.*, 2026, 42(6): 1436–1456.

SCHOLARSHIPS

Award for Graduate Students in Mathematics at Peking University 2025

Awarded by: Beijing International Center for Mathematical Research, Peking University

Scholarship of School of Mathematical Sciences 2026

Awarded by: School of Mathematical Sciences, Peking University